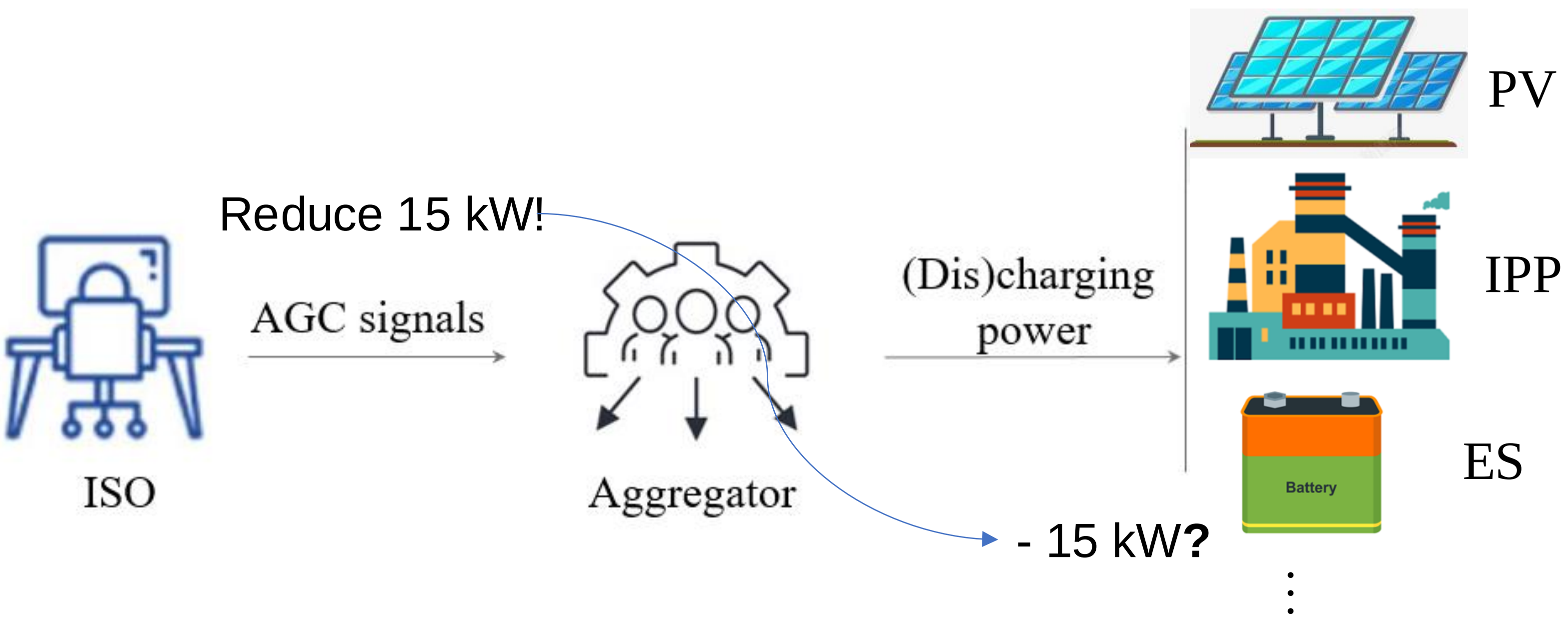
4-Optimization with Heterogeneous Resources

Motivation: when deploying regulation services, the VPP **disaggregates the requested power adjustment signals** in real time among its **internal heterogeneous resources**. Achieving optimal power disaggregation is challenging due to the **temporal coupling characteristics** of the resources. Existing research relies on **proportional disaggregation**, which is far from optimal.



Intuition: optimally allocate discharging power to resources with lower costs!

Can we always allocate discharging power to resources with lower costs?

4-Optimization with Heterogeneous Resources

Method (step1) : Establish a standardized VPP internal resource operation model, **using one set of models to describe the operation of various resources,** such as industrial production processes, air conditioning system, electric vehicle, energy storage.

$p_{t,s,i}=p^{\mathrm{dis}}_{t,s,i}-p^{\mathrm{ch}}_{t,s,i},\forall t,\forall s,\forall i$ $\underline{p}^{\mathrm{dis(ch)}}_{t,i}\le p^{\mathrm{dis(ch)}}_{t,s,i}\le \overline{p}^{\mathrm{dis(ch)}}_{t,i},\forall t,\forall s,\forall i$	Power limit
$\underline{e}_{t,i}\le e_{t,i}\le \overline{e}_{t,i},\forall t,\forall i$	State limit
$e_{t+1,i}=\theta_i e_{t,i}+\sum_s \pi_{t,s}\sum_j\left(\eta^{\mathrm{ch}}_{ij}p^{\mathrm{ch}}_{t,s,j}-\frac{1}{\eta^{\mathrm{dis}}_{ij}}p^{\mathrm{dis}}_{t,s,j}\right)\Delta t$ $+w_{t,i}\Delta t:\lambda^{e}_{t,i},\forall t,\forall i$	State change
$e^{\mathrm{init}}_{i}-e_{t,i}=0,t=1,\forall i$	Initial state
$\mathrm{Cost}_t=\sum_i\sum_{s\in[S]}\pi_{t,s}\left(\mathrm{Pr}^{\mathrm{dis}}_i p^{\mathrm{dis}}_{t,s,i}+\mathrm{Pr}^{\mathrm{ch}}_i p^{\mathrm{ch}}_{t,s,i}\right)\Delta t,\forall t$	Response cost

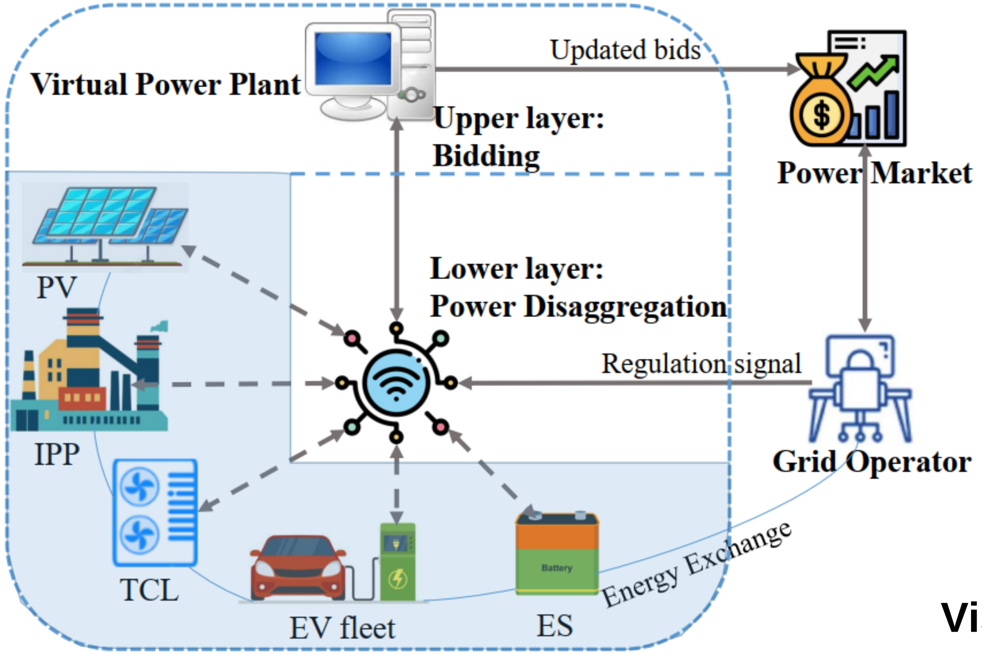
Standardized operating model for VPP internal resources

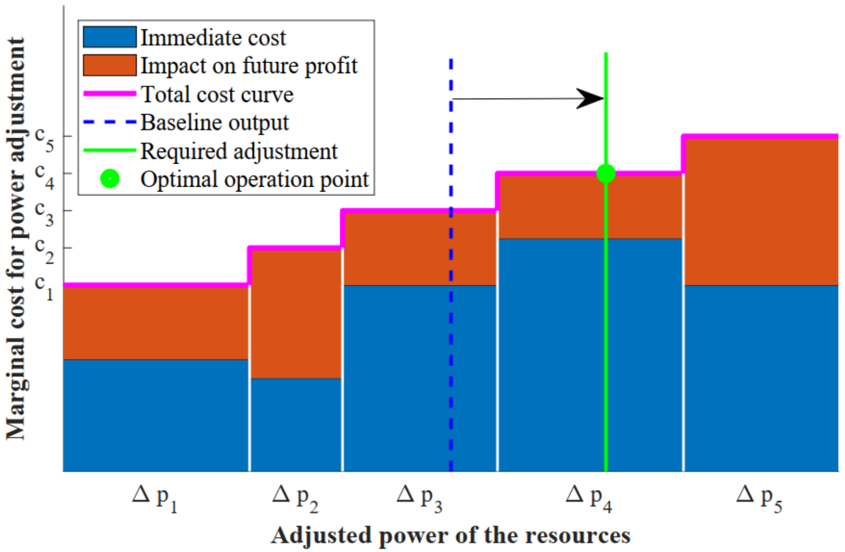
PV IPP TCL EV fleet ES

The modelling of IPP here is based on the **linearized state-task network (LSTN) we developed previously**. η^{ch}_{ij} is the inter-production efficiency between IPPs, which represents the change in material *i* caused by the energy consumed by production stage *j*.

4-Optimization with Heterogeneous Resources

Method (Step 2): Establish a multi-timescale optimization model, with the **upper layer** for **bid decision-making** and the **lower layer** responsible for **power disaggregation**. The optimization disaggregation strategy is to **prioritize the use of low-cost resources** while considering the coupling characteristics of time periods (i.e., the opportunity cost/ impact on future profit).





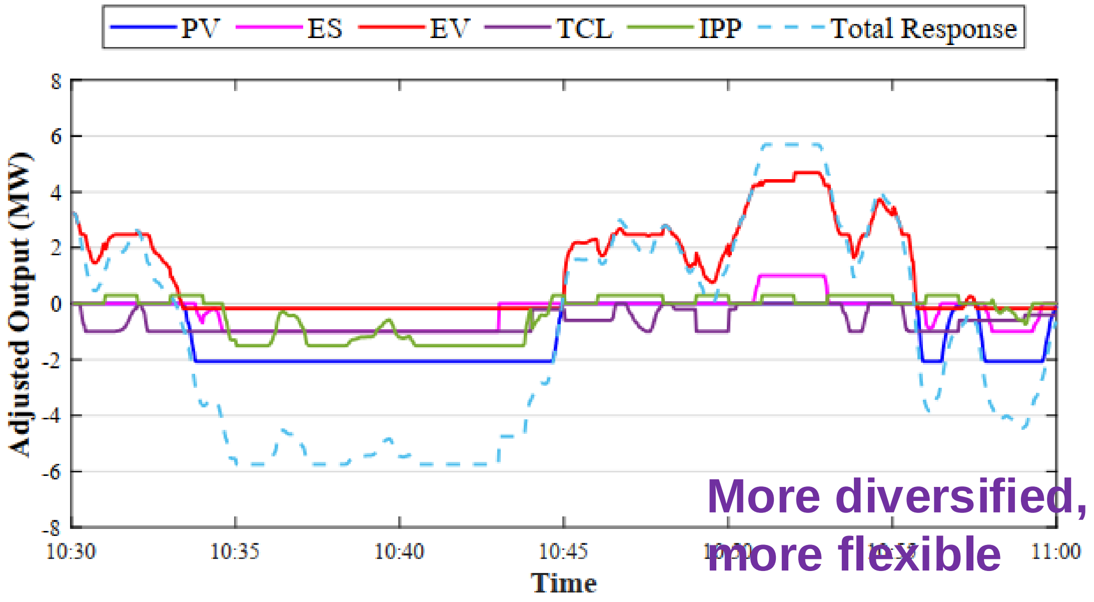
Visual representation of the Optimal Disaggregation algorithm

- Qixin Chen, Ruike Lyu, Hongye Guo, and Xiangbo Su. "Real-Time Operation Strategy of Virtual Power Plants With Optimal Power Disaggregation Among Heterogeneous Resources." Applied Energy 361 (2024): 122876.

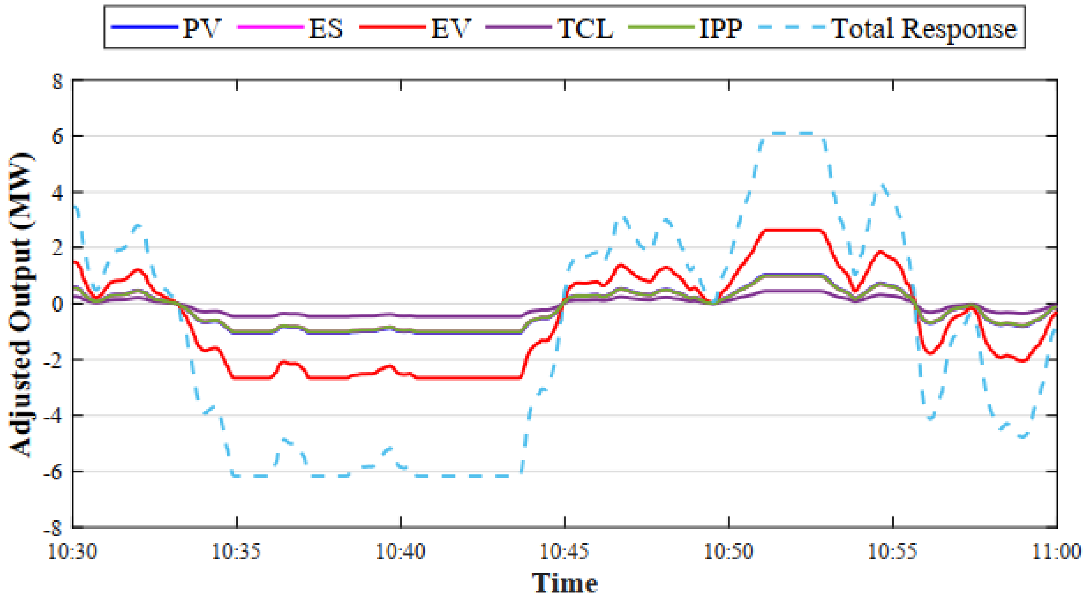
4-Optimization with Heterogeneous Resources

Case study: The VPP consists of **PV**, **ES**, a fleet of **EVs**, **TCLs**, and **IPP** resources.

We use the **real-time prices** of **energy** and **frequency regulation**of the PJM in July 2022 as boundary conditions.



(a) **Our method:**
power disaggregation optimally



(b) Compared method:
power disaggregated with the same proportion

4-Optimization with Heterogeneous Resources

Results: Achieving **real-time control optimization** of thousands of different types of resources (including industrial production processes) within **a calculation time of less than 1ms**, leading to **a 40% increase in VPP profit in energy & frequency regulation market** under typical scenarios.

calculation time

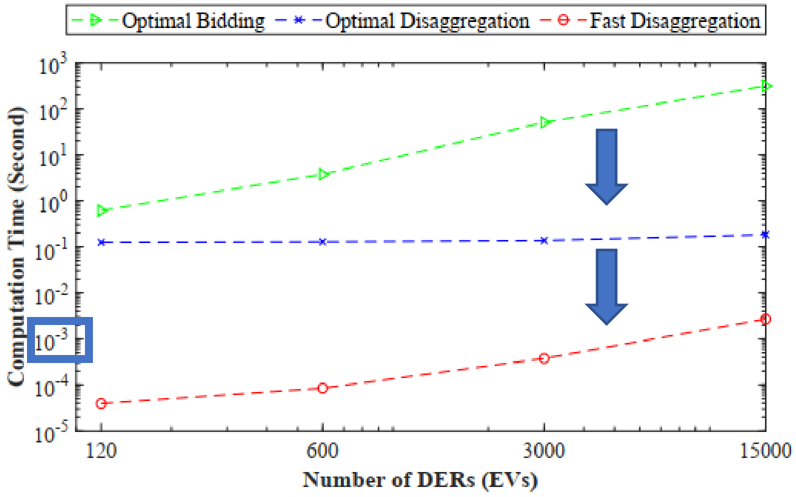


Table 1: Comparison of VPP Profits Using Different Methods

Bidding-Disaggregation Method	Income from the Market ($)	Operational Cost ($)	VPP Profit ($)
Proportional Disaggregation	1709 (0.0%)	1455 (0.0%)	254 (0.0)
Greedy Disaggregation	1316 (-23.0%)	1845 (+26.8%)	-529 (-782)
Optimal Disaggregation	**2622 (+53.3%)**	**1289 (-11.4%)**	**1333 (+1079)**

- Qixin Chen, Ruike Lyu, Hongye Guo, and Xiangbo Su. "Real-Time Operation Strategy of Virtual Power Plants With Optimal Power Disaggregation Among Heterogeneous Resources." Applied Energy 361 (2024): 122876.